
Intro to DB – Project 2

RECIPE DATABASE

Functional Dependencies

- $(\text{RecipeID}, \text{IngredientID}) \rightarrow \text{RecipeName}, \text{Notes}, \text{AcquiredFrom}, \text{IngredientID}, \text{Ingredient}, \text{Quantity}, \text{Technique}$
- $(\text{RecipeID}) \rightarrow \text{RecipeName}, \text{Notes}, \text{AcquiredFrom}$
- $(\text{IngredientID}) \rightarrow \text{Ingredient}$

Tables

- $\text{RECIPE_INGREDIENT}(\underline{\text{RecipeID}}, \underline{\text{IngredientID}}, \text{Quantity}, \text{Technique})$
- $\text{RECIPE}(\underline{\text{RecipeID}}, \text{RecipeName}, \text{Notes}, \text{AcquiredFrom})$
- $\text{INGREDIENT}(\underline{\text{IngredientID}}, \text{IngredientName})$

Notes

- Given FD's are non-negotiable, though there could be extra ones that we are meant to consider.
- There are a few things to consider: `acquired_from`, `quantity`, and `techniques`.
- `acquired_from` could have a look up table, though it wasn't required by the functional dependencies.
- `Quantity` is a weird one because it feels like you could either peel it off into a table and then separate the ideas into columns like 'units' and 'amount'. Though it still feels messy, and for the most part, `quantity` is uniquely described by `recipe` and `ingredient`. So I feel fine leaving it alone.
- `Technique` is another one that could get a look up table, but it also is a unique column that is determined by `recipe` and `ingredient`.
- All in all, I decided not to add any extra functional dependencies.

o